

LLaVA-DyMoE

Dynamic MoE with Drift-Aware Token Assignment for Continual Learning of LVLMs

Chongyang Zhao et al. · CVPR 2026

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Agenda

1. **Problem** — MCIT & MoE-CL, Routing Drift, Token's Dilemma
2. **Method** — TAG, RSR, Full Framework & Training Objective
3. **Validation** — Benchmark, SOTA Results, Ablation & Analysis

Problem: Continual Learning for MLLMs

MCIT Setting

- Sequential tasks $T_1, T_2, \dots, T_N$ arrive one-by-one
- Previous data **inaccessible** after each task ends
- Full retraining at MLLM scale is prohibitively expensive

MoE-based CL: The Idea

- Freeze old experts → preserve old knowledge
- Add new LoRA experts per task
→ parameter isolation

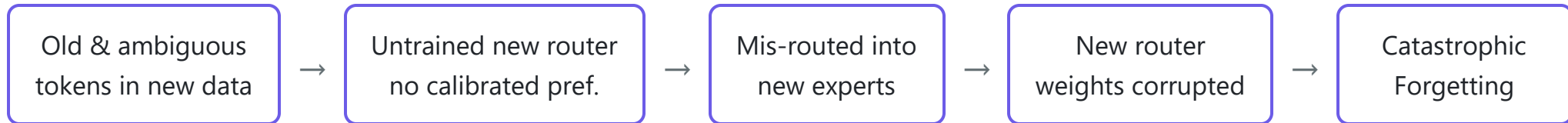
Why Existing MoE-CL Falls Short

Method	Core Limitation
Task-level routing	Requires task ID at inference
MoELoRA, ProgLoRA	Task-ID unavailable in practice
Token-level MoE	Routing drift is uncontrolled

MoE's defining strength is **token-level** flexibility — task-level routing discards it entirely

Core Problem: Routing Drift

Even with old experts **frozen**, the new router trains from scratch on unfiltered token mixtures — this alone is enough to destroy old-task knowledge.



freeze old params $\nRightarrow$ no routing corruption

Key Discovery: Token's Dilemma

New Tokens

Novel knowledge
No prior router preference
Natural → new experts
Safe to use

Old Tokens

Well-handled by old experts
Zero new-task value
If leaked → drift
Silent danger

Ambiguous Tokens

Boundary between old/new
Unstable routing
Alone: both tasks → 0%
Catastrophic trigger

The Core Dilemma: Only **new tokens** have learning value — yet **old & ambiguous tokens** impose catastrophic routing corruption costs if left uncontrolled.

Evidence: What Each Token Type Does



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Condition	New Task	Old Task	Implication
(a) New Tokens only	Strong ↑	Minimal loss	New tokens alone are sufficient
(b) Mask Old Tokens	Same	Same	Old tokens are expendable
(c) Ambiguous only	0%	0%	Ambiguous = forgetting trigger

Method: TAG — Token Assignment Guidance

Core Idea

Measure per-token routing ambiguity in **real-time** using the router's own logits —
no task labels, no historical data required.

Step 1: Group-wise Peak Logit

$$c_{old} = \max(\mathbf{s}_{t-1}), \quad c_{new} = \max(\mathbf{s}_{t,new})$$

Step 2: Relative Ambiguity Score

$$D_{rel} = \frac{|c_{new} - c_{old}|}{\max(|c_{new}|, |c_{old}|) + \epsilon}$$

Small $D_{rel} \rightarrow$ the router sees both groups as equally attractive $\rightarrow$ **ambiguous**

Step 3: Binary Gate

$$M_{new} = \mathbb{1}((c_{new} > c_{old}) \wedge (D_{rel} > \tau))$$

Tokens with $M_{new} = 0$ are hard-redirected to old experts.

Blocked tokens have new-expert logits set to $-\infty$:

$$s'_{t,i} = \begin{cases} s_{t,i} & \text{if } m_{t,i} = 1 \\ -\infty & \text{otherwise} \end{cases}$$

Method: RSR — Routing Score Regularization

Why TAG Alone Is Not Enough

TAG is a **hard binary mask** — it controls which expert group a token enters, but does not shape the soft routing weights within each group. RSR provides the complementary **continuous signal**.

Exclusivity Loss $\mathcal{L}_{exc}$

Push every token to commit exclusively to one group:

$$\mathcal{L}_{exc} = g_{old} \cdot g_{new}$$

Minimizing the product forces one term toward zero, preventing any token from being split across both groups.

Specialization Loss $\mathcal{L}_{spe}$

Ensure new experts are actually claimed by new tokens:

$$\tilde{g}_{old} = \max_{i \in \mathcal{S}_{t-1}} \{w_i\}, \quad y = 1 - \tilde{g}_{old}$$

$$\mathcal{L}_{spe} = -y \log g_{new} - (1 - y) \log(1 - g_{new})$$

When old experts cannot handle the token ($\tilde{g}_{old}$ low $\rightarrow y \approx 1$),

new expert weight g_{new} is maximized $\rightarrow$ **forced specialization**

TAG + RSR Together

TAG: hard gate $\rightarrow$ RSR: soft shaping

$\rightarrow$ Complete control over routing landscape

Method: Full Framework



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- **TAG mechanism** blocks ambiguous & old-domain tokens from entering new experts
- **RSR mechanism** (2 loss terms) enforces group separation & expert specialization
- **Inference:** Zero overhead — TAG + RSR active only during training

Method: Training Objective

$$\mathcal{L} = \mathcal{L}_{NTP} + \lambda \mathcal{L}_{aux} + \alpha (\mathcal{L}_{exc} + \mathcal{L}_{spe})$$

NTP Loss

Next-Token Prediction

Standard autoregressive cross-entropy for task learning

Aux Loss

Load Balancing

Uniform activation across new experts
Weight: $\lambda = 10^{-3}$

RSR Loss

Routing Regularization

Exclusivity + Specialization
Weight: $\alpha = 10^{-3}$

Experimental Setup: CoIN Benchmark

8 Sequential VQA Tasks (569k / 261k)

#	Task	Domain
1	SQA	Science reasoning
2	TextVQA	Text in images
3	ImageNet	Image classification
4	GQA	Multi-step reasoning
5	VizWiz	Assistive VQA
6	Grounding	Bounding box regression
7	VQAv2	Broad-domain VQA
8	OCR-VQA	Text-logic reasoning

Evaluation Metrics

Metric	Meaning
MFN	Avg. accuracy after all tasks
MAA	Avg. across all training stages
BWT	Forgetting rate (closer to 0 = better)

Backbone & Config

- LLaVA-v1.5-7B / 13B (Vicuna + CLIP ViT-L/14)
- Base frozen; only LoRA experts & router trained
- 16 rank-4 LoRA experts per task, Top-16 routing
- $\tau = 20\%$, $\lambda = 10^{-3}$, $\alpha = 10^{-3}$

Results: SOTA on CoIN



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	IncMoELoRA (Baseline)	LLaVA-DyMoE (Ours)	Gain
MFN	49.68%	57.03%	+7.35%p
BWT	-16.67%	-4.67%	+12.00%p

Outperforms EWC, LwF, O-LoRA, and all MoE-CL baselines on both plasticity (MFN) and stability (BWT)

Ablation: Component Contribution

Incremental Gains

Config	MFN	BWT
Base (IncMoELoRA)	49.68%	-16.67%
+ TAG	54.44%	-7.04%
+ $\mathcal{L}_{exc}$	55.18%	—
+ $\mathcal{L}_{spe}$ (Full)	57.03%	-4.67%

Reading the Results

- **TAG** delivers the largest single gain: fixing the root cause (mis-routing) matters most
- $\mathcal{L}_{exc}$ tightens group separation — TAG's hard gate, now softly enforced
- $\mathcal{L}_{spe}$ maximizes new expert utility — completing the specialization loop



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Ablation: τ Threshold Sensitivity



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τ Controls the Plasticity–Stability Trade-off

- **Too low (10%):** Ambiguous tokens pass through → routing drift persists → forgetting
- **Optimal (20%):** MFN 57.03%, BWT −4.67% — best balance
- **Too high (30%+):** Even clean new tokens get blocked → learning degraded (MFN ↓)

τ is the **ambiguity detection threshold** —
not a forgetting penalty, but a filter for which tokens are
safe to send to new experts

Analysis: Expert Activation Patterns



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X-axis = Layer index (0–32)
Y-axis = Expert group (1–8)
Circle size = Cumulative routing weight

Key Findings

- **Diverse layer-wise patterns** → genuine domain specialization per depth
- **Early expert groups stable** → ID 1–4 remain activated after 8 tasks
- **No routing corruption** → TAG & RSR successfully preserve old routing policy

Qualitative: VQA Predictions



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Conclusion

Core Contributions

1. **Token's Dilemma** — identifies that routing drift originates from specific token types, not from expert-level interference
2. **TAG** — real-time, label-free token filtering via router logit ambiguity; stops drift at the source
3. **RSR** — $\mathcal{L}_{exc} + \mathcal{L}_{spe}$: soft regularization for crisp group separation and expert specialization
4. **SOTA on CoIN** — MFN +7.35%p, BWT +12.00%p over the strongest MoE-CL baseline

Strengths

- Token-level control — no task ID required at inference
- Training-only overhead — zero inference cost
- Plug-and-play with existing CL frameworks

Limitations

- τ requires tuning per setup
- Expert memory grows linearly with task count
- Evaluation limited to MCIT / VQA domains

Thank You

Questions?